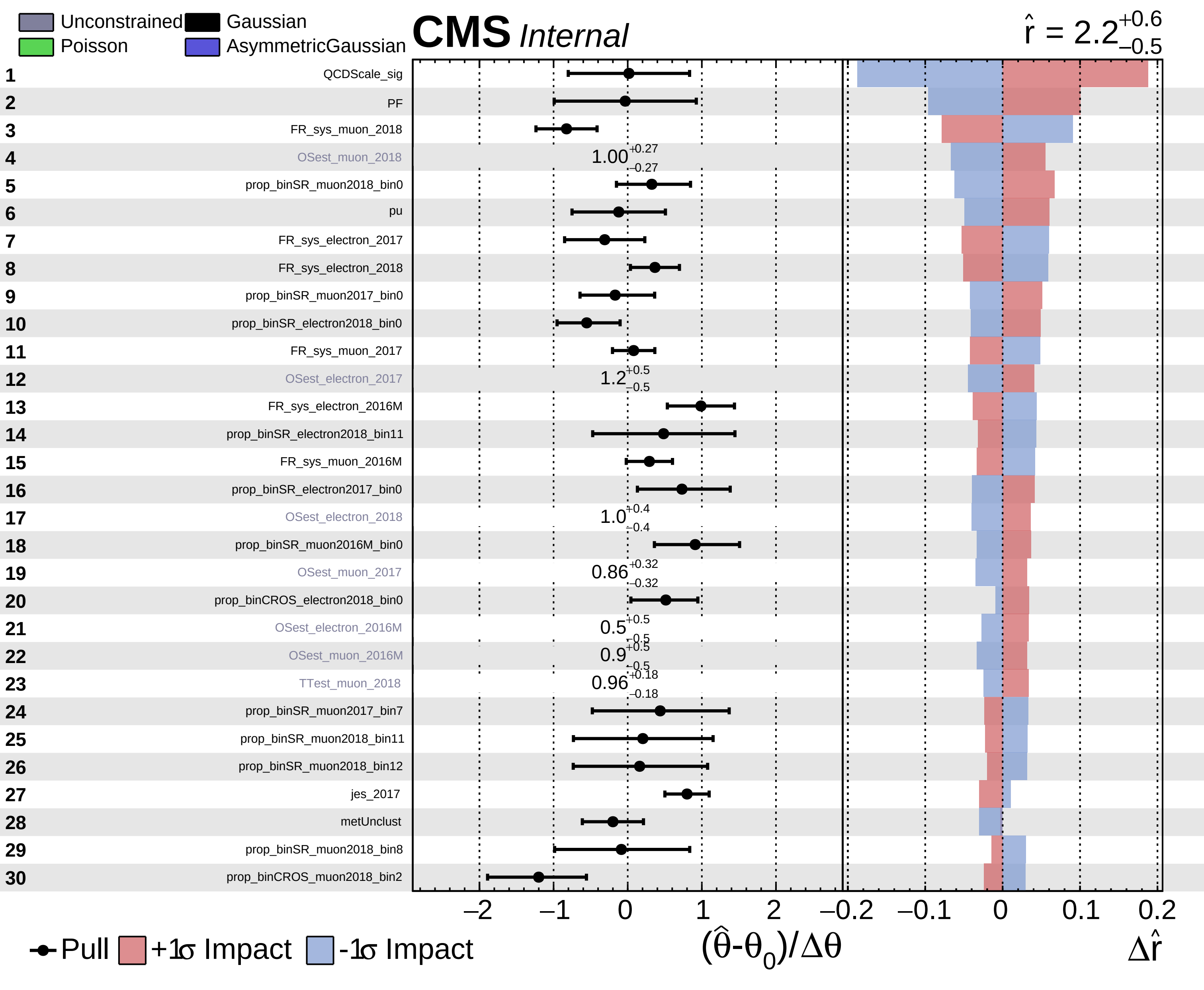
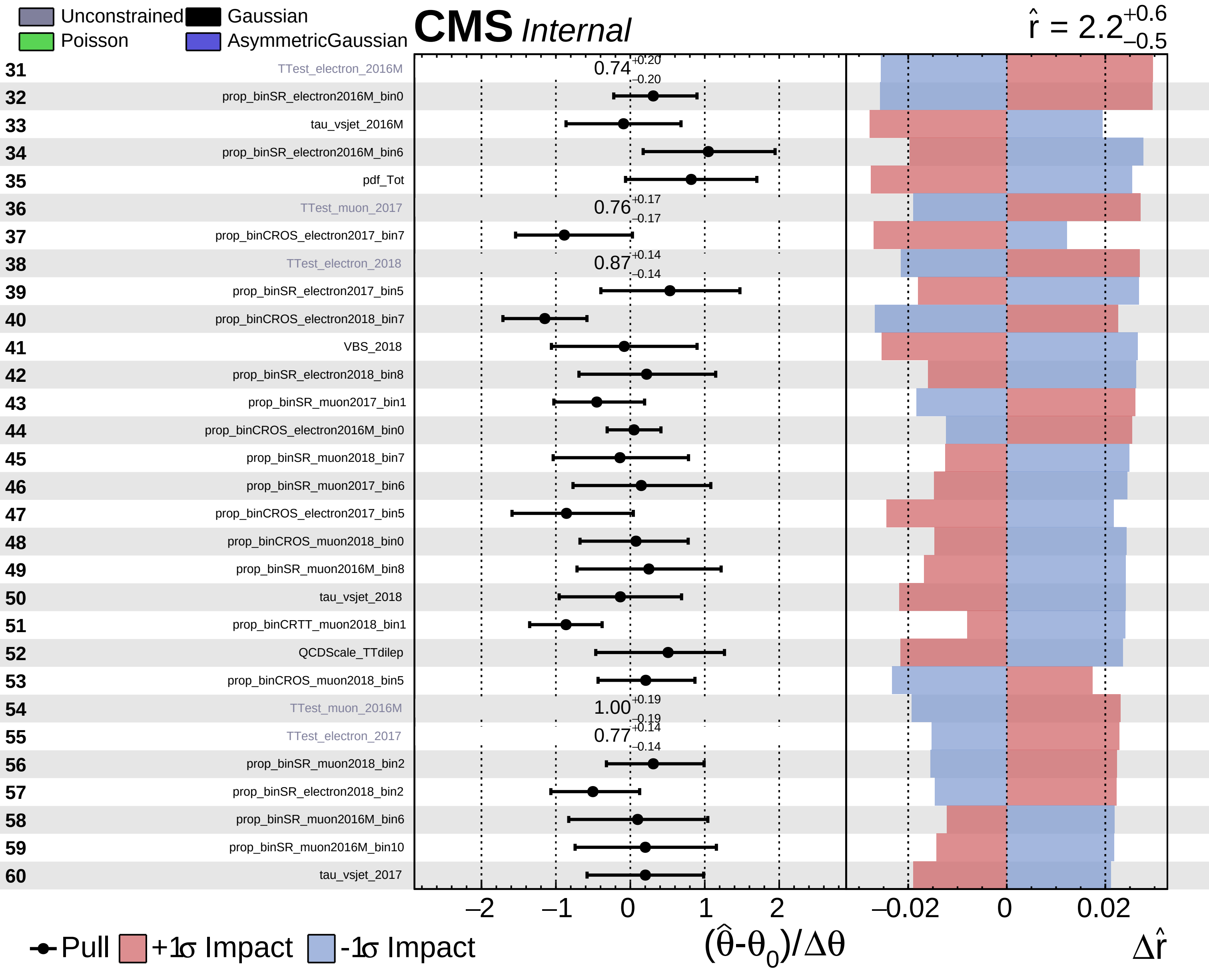
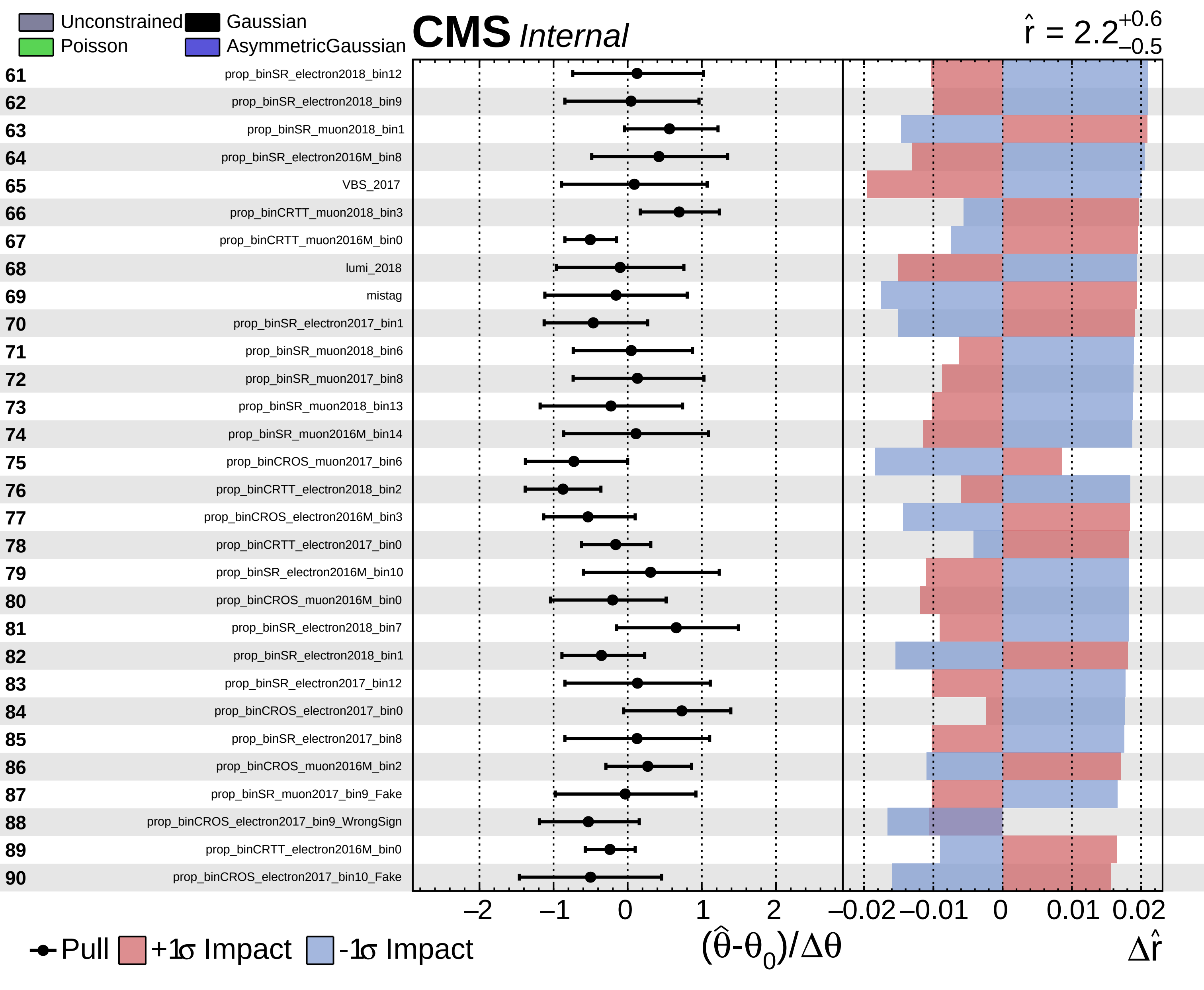
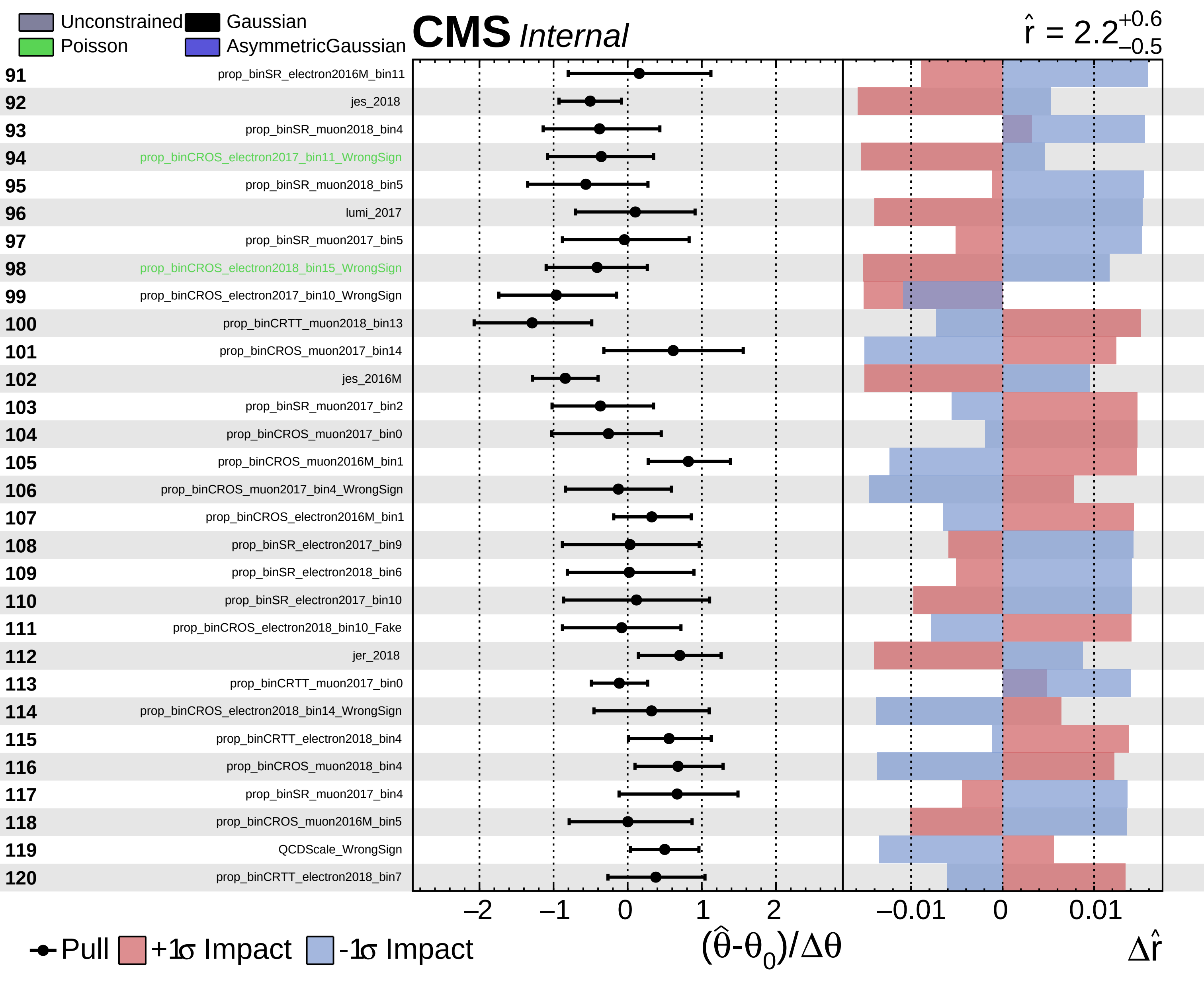


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.2^{+0.6}_{-0.5}$
1 QCDScale_sig
2 PF
3 FR_sys_muon_2018
4 OSest_muon_2018
$1.00^{+0.27}_{-0.27}$
5 prop_binSR_muon2018_bin0
6 pu
7 FR_sys_electron_2017
8 FR_sys_electron_2018
9 prop_binSR_muon2017_bin0
10 prop_binSR_electron2018_bin0
11 FR_sys_muon_2017
12 OSest_electron_2017
$1.2^{+0.5}_{-0.5}$
13 FR_sys_electron_2016M
14 prop_binSR_electron2018_bin11
15 FR_sys_muon_2016M
16 prop_binSR_electron2017_bin0
17 OSest_electron_2018
$1.0^{+0.4}_{-0.4}$
18 prop_binSR_muon2016M_bin0
19 OSest_muon_2017
$0.86^{+0.32}_{-0.32}$
20 prop_binCROS_electron2018_bin0
21 OSest_electron_2016M
$0.5^{+0.5}_{-0.5}$
22 OSest_muon_2016M
$0.9^{+0.5}_{-0.5}$
23 TTest_muon_2018
$0.96^{+0.18}_{-0.18}$
24 prop_binSR_muon2017_bin7
25 prop_binSR_muon2018_bin11
26 prop_binSR_muon2018_bin12
27 jes_2017
28 metUnclust
29 prop_binSR_muon2018_bin8
30 prop_binCROS_muon2018_bin2
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

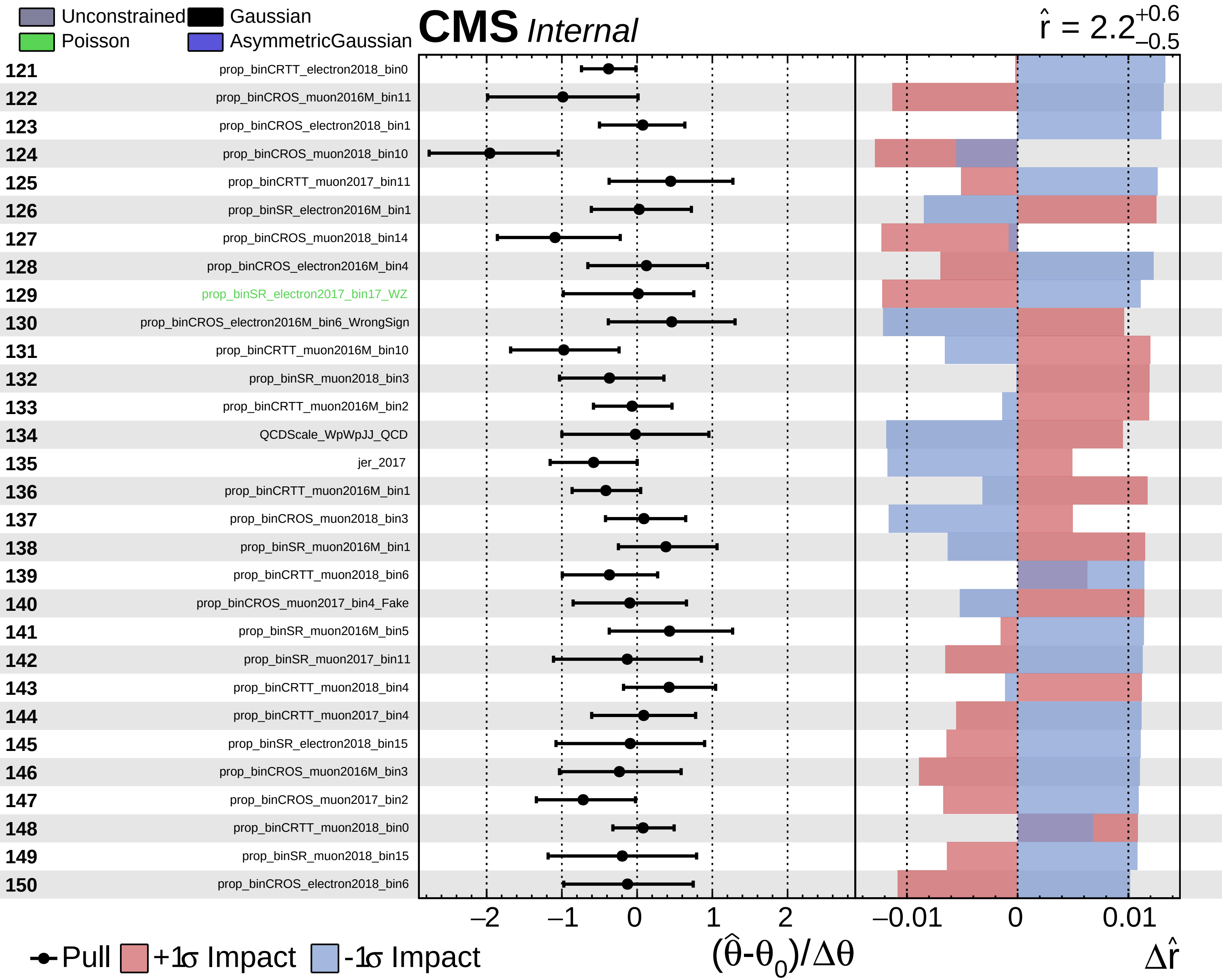

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.2^{+0.6}_{-0.5}$
31 TTest_electron_2016M 0.74 +0.20 −0.20
32 prop_binSR_electron2016M_bin0
33 tau_vsjet_2016M
34 prop_binSR_electron2016M_bin6
35 pdf_Tot
36 TTest_muon_2017 0.76 +0.17 −0.17
37 prop_binCROS_electron2017_bin7
38 TTest_electron_2018 0.87 +0.14 −0.14
39 prop_binSR_electron2017_bin5
40 prop_binCROS_electron2018_bin7
41 VBS_2018
42 prop_binSR_electron2018_bin8
43 prop_binSR_muon2017_bin1
44 prop_binCROS_electron2016M_bin0
45 prop_binSR_muon2018_bin7
46 prop_binSR_muon2017_bin6
47 prop_binCROS_electron2017_bin5
48 prop_binCROS_muon2018_bin0
49 prop_binSR_muon2016M_bin8
50 tau_vsjet_2018
51 prop_binCRTT_muon2018_bin1
52 QCDScale_TTdilep
53 prop_binCROS_muon2018_bin5
54 TTest_muon_2016M 1.00 +0.19 −0.19
55 TTest_electron_2017 0.77 +0.14 −0.14
56 prop_binSR_muon2018_bin2
57 prop_binSR_electron2018_bin2
58 prop_binSR_muon2016M_bin6
59 prop_binSR_muon2016M_bin10
60 tau_vsjet_2017
−2 −1 0 1 2
−0.02 0 0.02
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

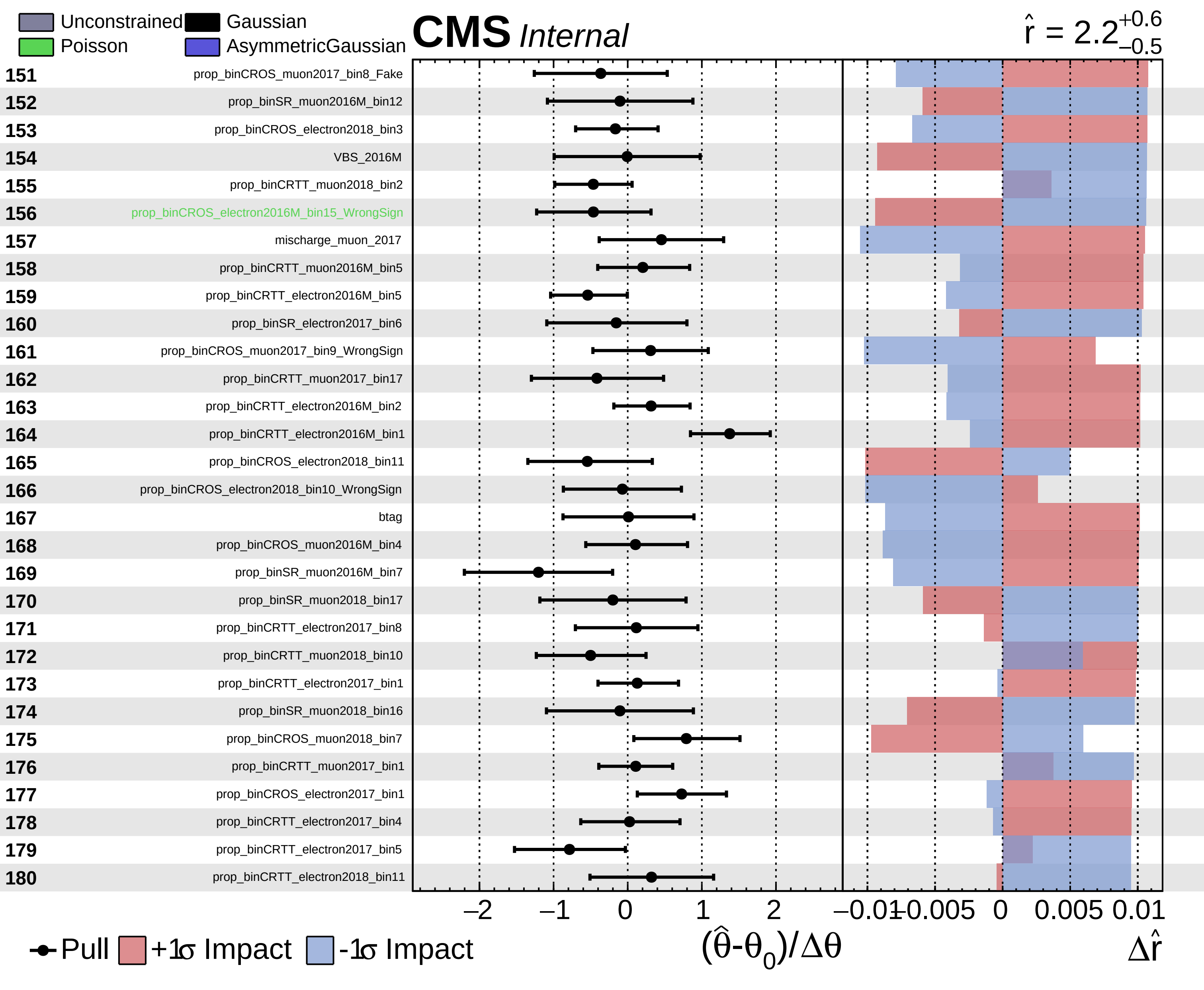


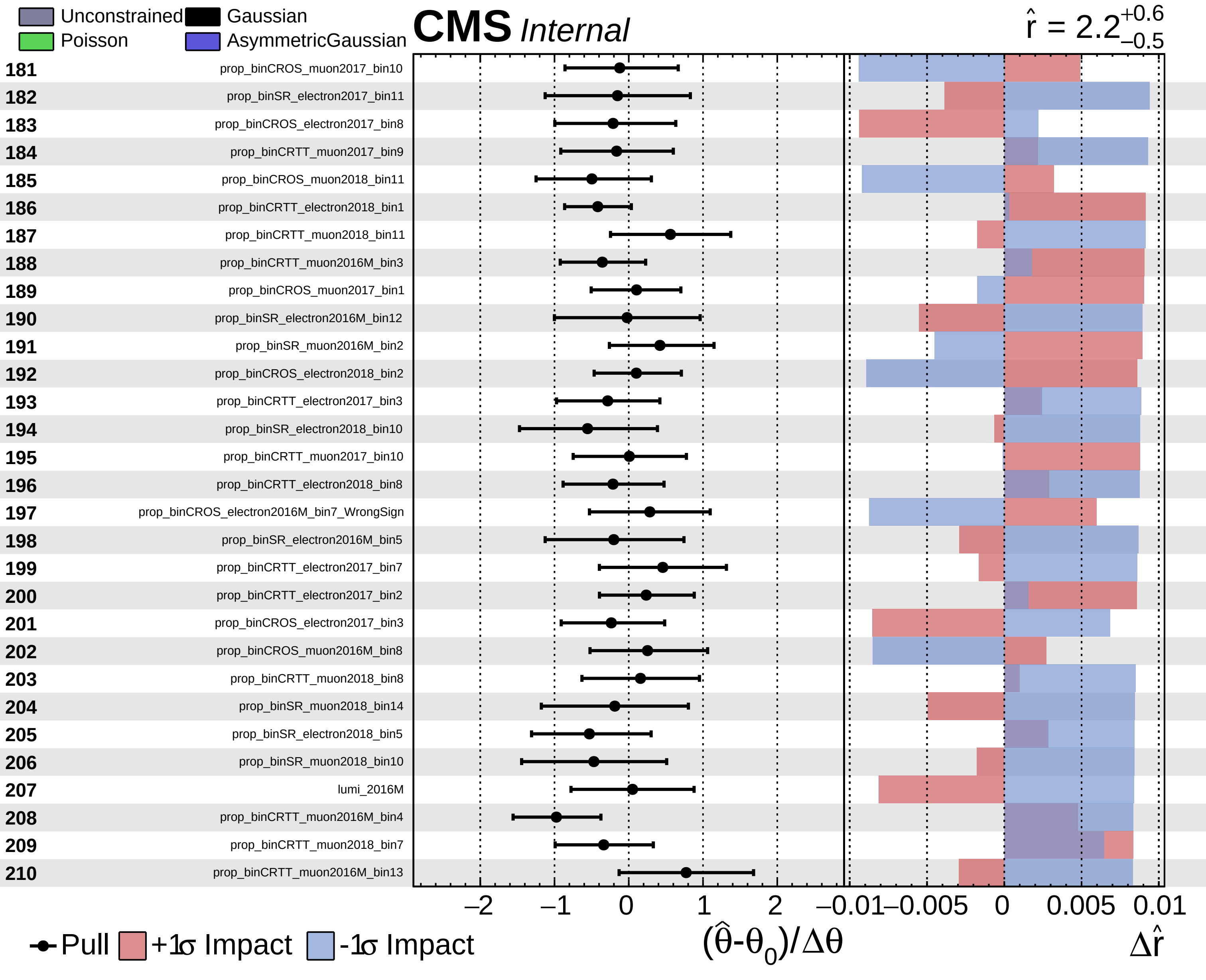
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.2^{+0.6}_{-0.5}$
61 prop_binSR_electron2018_bin12
62 prop_binSR_electron2018_bin9
63 prop_binSR_muon2018_bin1
64 prop_binSR_electron2016M_bin8
65 VBS_2017
66 prop_binCRTT_muon2018_bin3
67 prop_binCRTT_muon2016M_bin0
68 lumi_2018
69 mistag
70 prop_binSR_electron2017_bin1
71 prop_binSR_muon2018_bin6
72 prop_binSR_muon2017_bin8
73 prop_binSR_muon2018_bin13
74 prop_binSR_muon2016M_bin14
75 prop_binCROS_muon2017_bin6
76 prop_binCRTT_electron2018_bin2
77 prop_binCROS_electron2016M_bin3
78 prop_binCRTT_electron2017_bin0
79 prop_binSR_electron2016M_bin10
80 prop_binCROS_muon2016M_bin0
81 prop_binSR_electron2018_bin7
82 prop_binSR_electron2018_bin1
83 prop_binSR_electron2017_bin12
84 prop_binCROS_electron2017_bin0
85 prop_binSR_electron2017_bin8
86 prop_binCROS_muon2016M_bin2
87 prop_binSR_muon2017_bin9_Fake
88 prop_binCROS_electron2017_bin9_WrongSign
89 prop_binCRTT_electron2016M_bin0
90 prop_binCROS_electron2017_bin10_Fake
−2 −1 0 1 2
−0.02 −0.01 0 0.01 0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



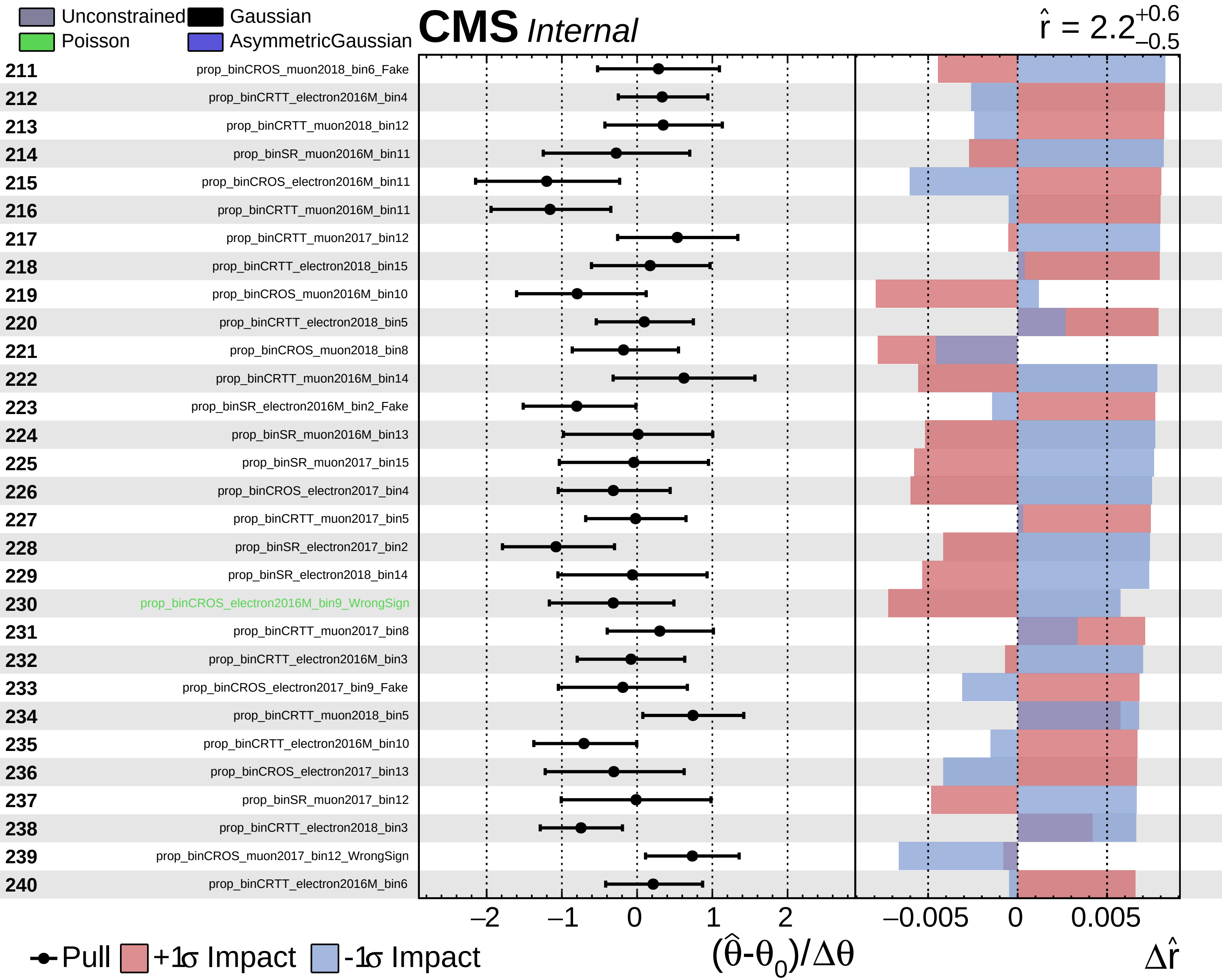
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.2^{+0.6}_{-0.5}$
91 prop_binSR_electron2016M_bin11
92 jes_2018
93 prop_binSR_muon2018_bin4
94 prop_binCROS_electron2017_bin11_WrongSign
95 prop_binSR_muon2018_bin5
96 lumi_2017
97 prop_binSR_muon2017_bin5
98 prop_binCROS_electron2018_bin15_WrongSign
99 prop_binCROS_electron2017_bin10_WrongSign
100 prop_binCRTT_muon2018_bin13
101 prop_binCROS_muon2017_bin14
102 jes_2016M
103 prop_binSR_muon2017_bin2
104 prop_binCROS_muon2017_bin0
105 prop_binCROS_muon2016M_bin1
106 prop_binCROS_muon2017_bin4_WrongSign
107 prop_binCROS_electron2016M_bin1
108 prop_binSR_electron2017_bin9
109 prop_binSR_electron2018_bin6
110 prop_binSR_electron2017_bin10
111 prop_binCROS_electron2018_bin10_Fake
112 jer_2018
113 prop_binCRTT_muon2017_bin0
114 prop_binCROS_electron2018_bin14_WrongSign
115 prop_binCRTT_electron2018_bin4
116 prop_binCROS_muon2018_bin4
117 prop_binSR_muon2017_bin4
118 prop_binCROS_muon2016M_bin5
119 QCDScale_WrongSign
120 prop_binCRTT_electron2018_bin7
−2
−1
0
1
2
−0.01
0
0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.2^{+0.6}_{-0.5}$
121 prop_binCRTT_electron2018_bin0
122 prop_binCROS_muon2016M_bin11
123 prop_binCROS_electron2018_bin1
124 prop_binCROS_muon2018_bin10
125 prop_binCRTT_muon2017_bin11
126 prop_binSR_electron2016M_bin1
127 prop_binCROS_muon2018_bin14
128 prop_binCROS_electron2016M_bin4
129 prop_binSR_electron2017_bin17_WZ
130 prop_binCROS_electron2016M_bin6_WrongSign
131 prop_binCRTT_muon2016M_bin10
132 prop_binSR_muon2018_bin3
133 prop_binCRTT_muon2016M_bin2
134 QCDScale_WpWpJJ_QCD
135 jer_2017
136 prop_binCRTT_muon2016M_bin1
137 prop_binCROS_muon2018_bin3
138 prop_binSR_muon2016M_bin1
139 prop_binCRTT_muon2018_bin6
140 prop_binCROS_muon2017_bin4_Fake
141 prop_binSR_muon2016M_bin5
142 prop_binSR_muon2017_bin11
143 prop_binCRTT_muon2018_bin4
144 prop_binCRTT_muon2017_bin4
145 prop_binSR_electron2018_bin15
146 prop_binCROS_muon2016M_bin3
147 prop_binCROS_muon2017_bin2
148 prop_binCRTT_muon2018_bin0
149 prop_binSR_muon2018_bin15
150 prop_binCROS_electron2018_bin6
−2 −1 0 1 2
−0.01 0 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

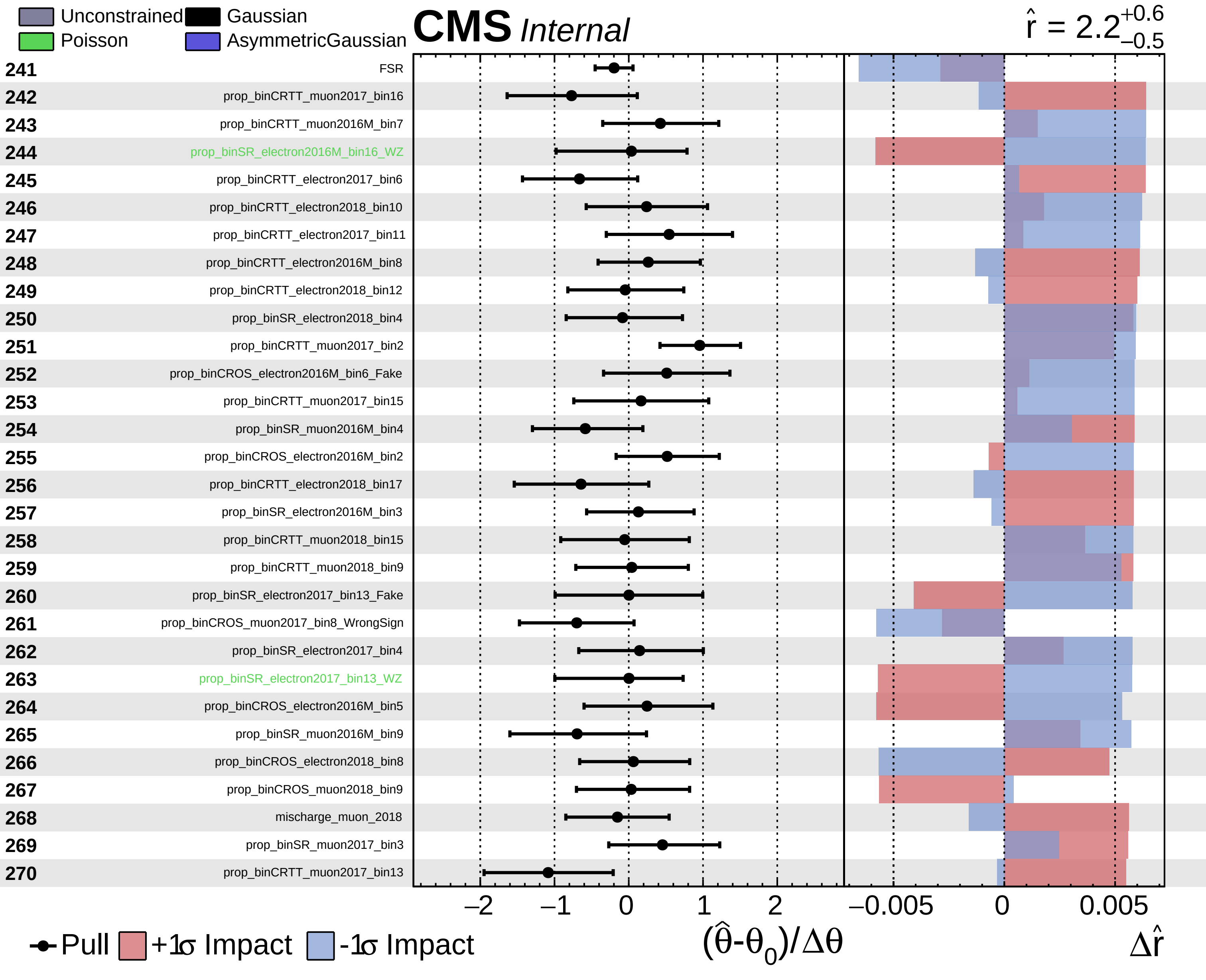

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.2^{+0.6}_{-0.5}$
151 prop_binCROS_muon2017_bin8_Fake
152 prop_binSR_muon2016M_bin12
153 prop_binCROS_electron2018_bin3
154 VBS_2016M
155 prop_binCRTT_muon2018_bin2
156 prop_binCROS_electron2016M_bin15_WrongSign
157 mischarge_muon_2017
158 prop_binCRTT_muon2016M_bin5
159 prop_binCRTT_electron2016M_bin5
160 prop_binSR_electron2017_bin6
161 prop_binCROS_muon2017_bin9_WrongSign
162 prop_binCRTT_muon2017_bin17
163 prop_binCRTT_electron2016M_bin2
164 prop_binCRTT_electron2016M_bin1
165 prop_binCROS_electron2018_bin11
166 prop_binCROS_electron2018_bin10_WrongSign
167 btag
168 prop_binCROS_muon2016M_bin4
169 prop_binSR_muon2016M_bin7
170 prop_binSR_muon2018_bin17
171 prop_binCRTT_electron2017_bin8
172 prop_binCRTT_muon2018_bin10
173 prop_binCRTT_electron2017_bin1
174 prop_binSR_muon2018_bin16
175 prop_binCROS_muon2018_bin7
176 prop_binCRTT_muon2017_bin1
177 prop_binCROS_electron2017_bin1
178 prop_binCRTT_electron2017_bin4
179 prop_binCRTT_electron2017_bin5
180 prop_binCRTT_electron2018_bin11
−2 −1 0 1 2
−0.01 −0.005 0 0.005 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



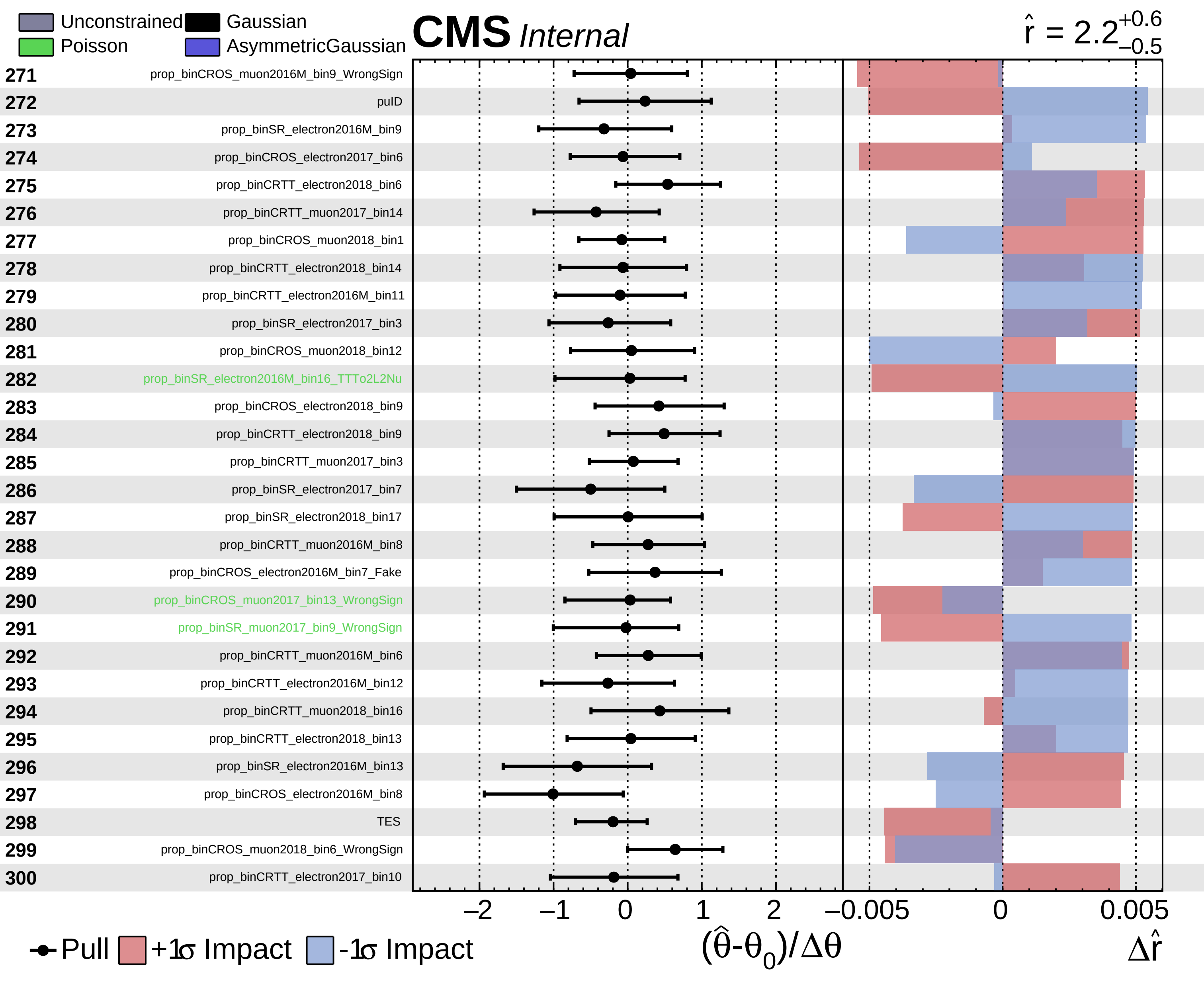
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.2^{+0.6}_{-0.5}$
181 prop_binCROS_muon2017_bin10
182 prop_binSR_electron2017_bin11
183 prop_binCROS_electron2017_bin8
184 prop_binCRTT_muon2017_bin9
185 prop_binCROS_muon2018_bin11
186 prop_binCRTT_electron2018_bin1
187 prop_binCRTT_muon2018_bin11
188 prop_binCRTT_muon2016M_bin3
189 prop_binCROS_muon2017_bin1
190 prop_binSR_electron2016M_bin12
191 prop_binSR_muon2016M_bin2
192 prop_binCROS_electron2018_bin2
193 prop_binCRTT_electron2017_bin3
194 prop_binSR_electron2018_bin10
195 prop_binCRTT_muon2017_bin10
196 prop_binCRTT_electron2018_bin8
197 prop_binCROS_electron2016M_bin7_WrongSign
198 prop_binSR_electron2016M_bin5
199 prop_binCRTT_electron2017_bin7
200 prop_binCRTT_electron2017_bin2
201 prop_binCROS_electron2017_bin3
202 prop_binCROS_muon2016M_bin8
203 prop_binCRTT_muon2018_bin8
204 prop_binSR_muon2018_bin14
205 prop_binSR_electron2018_bin5
206 prop_binSR_muon2018_bin10
207 lumi_2016M
208 prop_binCRTT_muon2016M_bin4
209 prop_binCRTT_muon2018_bin7
210 prop_binCRTT_muon2016M_bin13
−2 −1 0 1 2
−0.01 −0.005 0 0.005 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



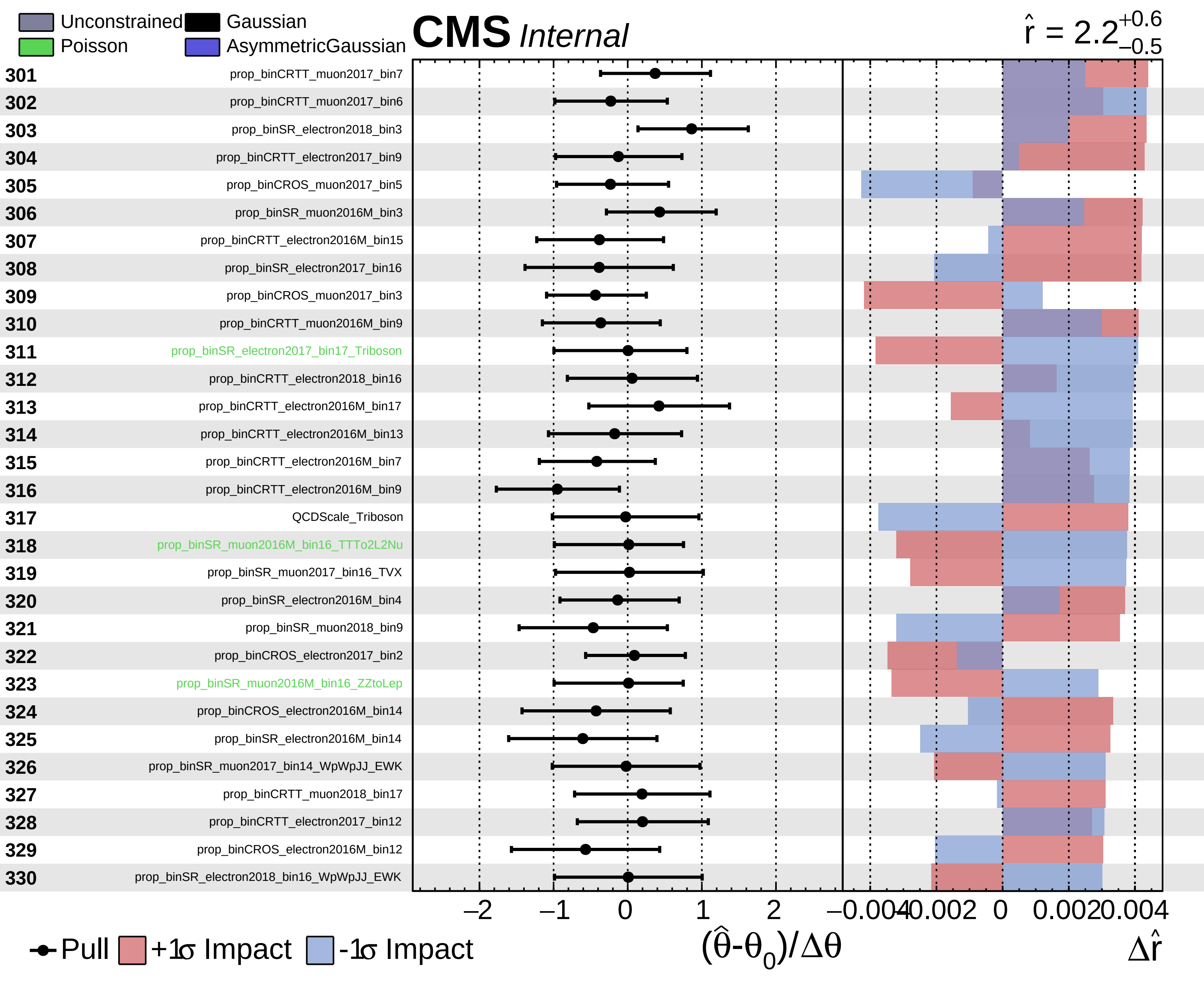
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.2^{+0.6}_{-0.5}$
211 prop_binCROS_muon2018_bin6_Fake
212 prop_binCRTT_electron2016M_bin4
213 prop_binCRTT_muon2018_bin12
214 prop_binSR_muon2016M_bin11
215 prop_binCROS_electron2016M_bin11
216 prop_binCRTT_muon2016M_bin11
217 prop_binCRTT_muon2017_bin12
218 prop_binCRTT_electron2018_bin15
219 prop_binCROS_muon2016M_bin10
220 prop_binCRTT_electron2018_bin5
221 prop_binCROS_muon2018_bin8
222 prop_binCRTT_muon2016M_bin14
223 prop_binSR_electron2016M_bin2_Fake
224 prop_binSR_muon2016M_bin13
225 prop_binSR_muon2017_bin15
226 prop_binCROS_electron2017_bin4
227 prop_binCRTT_muon2017_bin5
228 prop_binSR_electron2017_bin2
229 prop_binSR_electron2018_bin14
230 prop_binCROS_electron2016M_bin9_WrongSign
231 prop_binCRTT_muon2017_bin8
232 prop_binCRTT_electron2016M_bin3
233 prop_binCROS_electron2017_bin9_Fake
234 prop_binCRTT_muon2018_bin5
235 prop_binCRTT_electron2016M_bin10
236 prop_binCROS_electron2017_bin13
237 prop_binSR_muon2017_bin12
238 prop_binCRTT_electron2018_bin3
239 prop_binCROS_muon2017_bin12_WrongSign
240 prop_binCRTT_electron2016M_bin6
−2 −1 0 1 2
−0.005 0 0.005
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



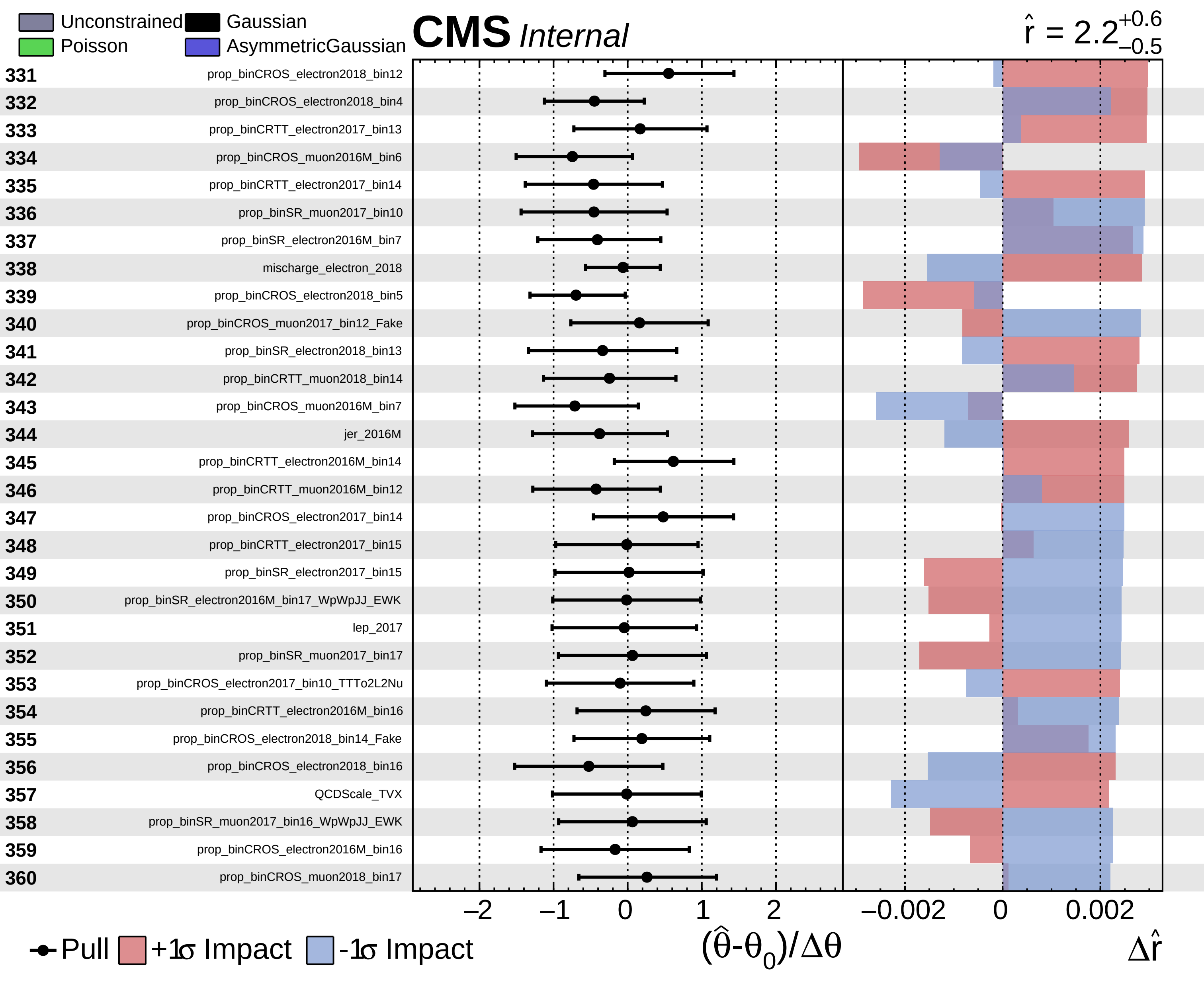
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.2^{+0.6}_{-0.5}$
241 FSR
242 prop_binCRTT_muon2017_bin16
243 prop_binCRTT_muon2016M_bin7
244 prop_binSR_electron2016M_bin16_WZ
245 prop_binCRTT_electron2017_bin6
246 prop_binCRTT_electron2018_bin10
247 prop_binCRTT_electron2017_bin11
248 prop_binCRTT_electron2016M_bin8
249 prop_binCRTT_electron2018_bin12
250 prop_binSR_electron2018_bin4
251 prop_binCRTT_muon2017_bin2
252 prop_binCROS_electron2016M_bin6_Fake
253 prop_binCRTT_muon2017_bin15
254 prop_binSR_muon2016M_bin4
255 prop_binCROS_electron2016M_bin2
256 prop_binCRTT_electron2018_bin17
257 prop_binSR_electron2016M_bin3
258 prop_binCRTT_muon2018_bin15
259 prop_binCRTT_muon2018_bin9
260 prop_binSR_electron2017_bin13_Fake
261 prop_binCROS_muon2017_bin8_WrongSign
262 prop_binSR_electron2017_bin4
263 prop_binSR_electron2017_bin13_WZ
264 prop_binCROS_electron2016M_bin5
265 prop_binSR_muon2016M_bin9
266 prop_binCROS_electron2018_bin8
267 prop_binCROS_muon2018_bin9
268 mischarge_muon_2018
269 prop_binSR_muon2017_bin3
270 prop_binCRTT_muon2017_bin13
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



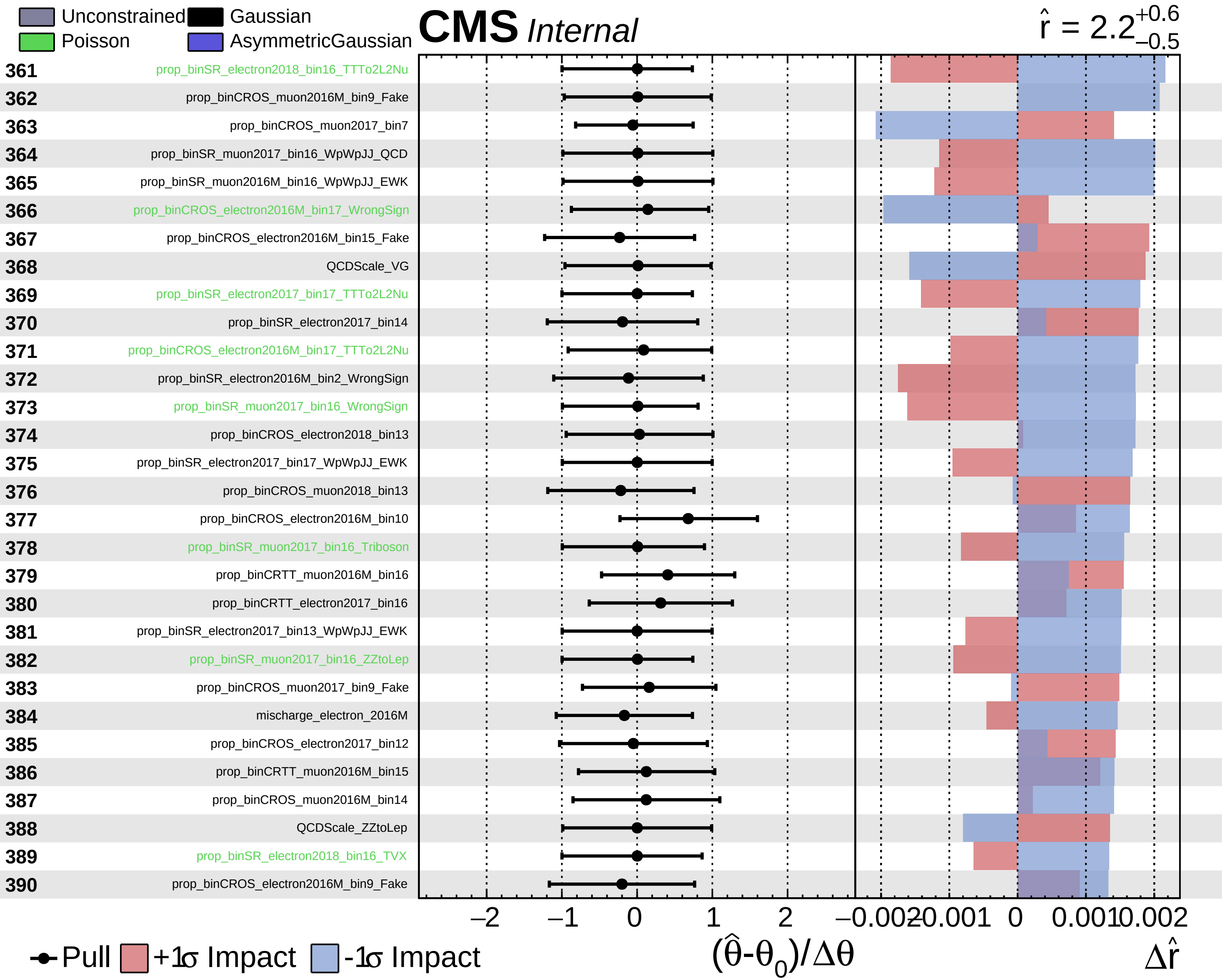
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.2^{+0.6}_{-0.5}$
271 prop_binCROS_muon2016M_bin9_WrongSign
272 puID
273 prop_binSR_electron2016M_bin9
274 prop_binCROS_electron2017_bin6
275 prop_binCRTT_electron2018_bin6
276 prop_binCRTT_muon2017_bin14
277 prop_binCROS_muon2018_bin1
278 prop_binCRTT_electron2018_bin14
279 prop_binCRTT_electron2016M_bin11
280 prop_binSR_electron2017_bin3
281 prop_binCROS_muon2018_bin12
282 prop_binSR_electron2016M_bin16_TTTo2L2Nu
283 prop_binCROS_electron2018_bin9
284 prop_binCRTT_electron2018_bin9
285 prop_binCRTT_muon2017_bin3
286 prop_binSR_electron2017_bin7
287 prop_binSR_electron2018_bin17
288 prop_binCRTT_muon2016M_bin8
289 prop_binCROS_electron2016M_bin7_Fake
290 prop_binCROS_muon2017_bin13_WrongSign
291 prop_binSR_muon2017_bin9_WrongSign
292 prop_binCRTT_muon2016M_bin6
293 prop_binCRTT_electron2016M_bin12
294 prop_binCRTT_muon2018_bin16
295 prop_binCRTT_electron2018_bin13
296 prop_binSR_electron2016M_bin13
297 prop_binCROS_electron2016M_bin8
298 TES
299 prop_binCROS_muon2018_bin6_WrongSign
300 prop_binCRTT_electron2017_bin10
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



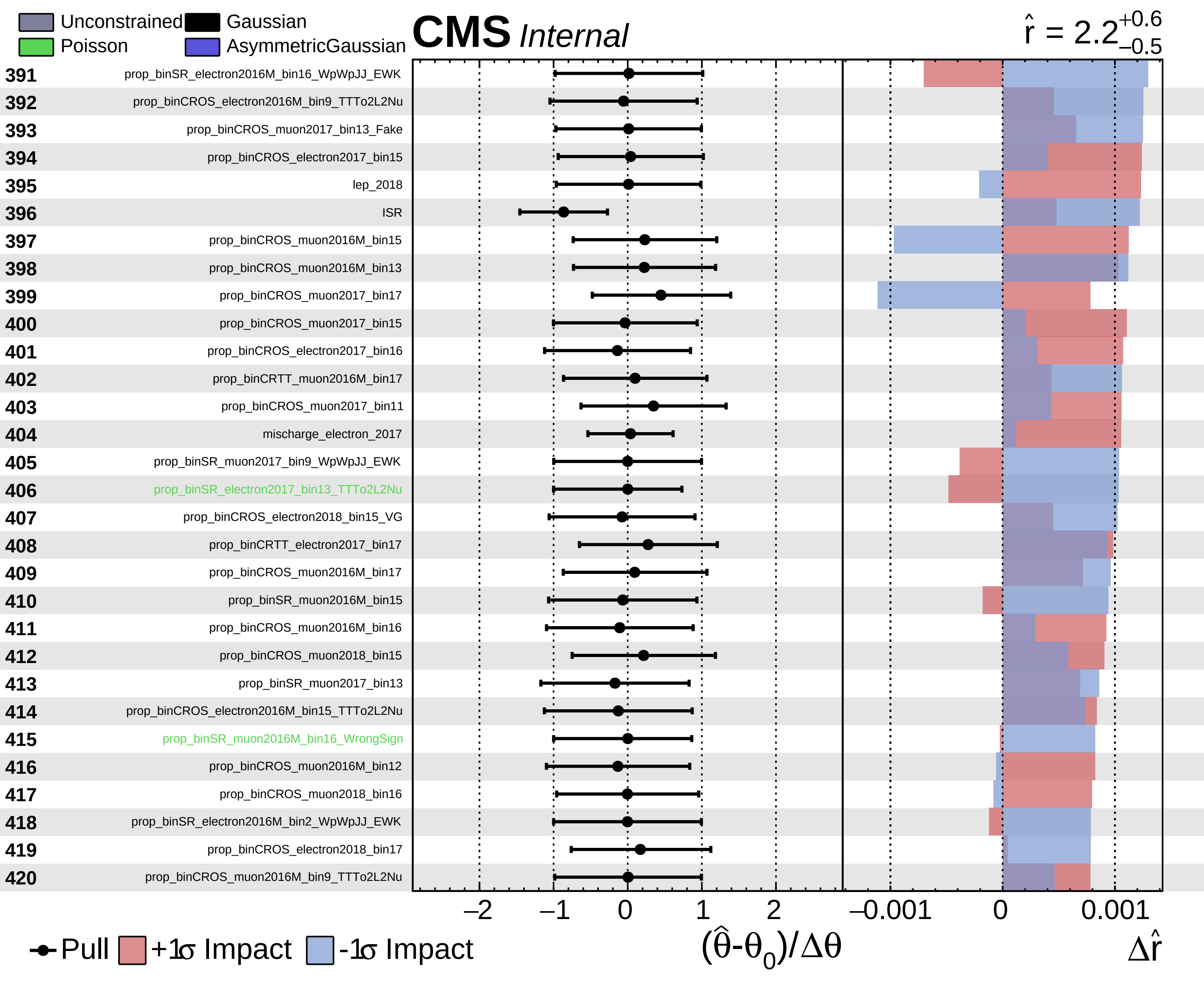
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.2^{+0.6}_{-0.5}$
301 prop_binCRTT_muon2017_bin7
302 prop_binCRTT_muon2017_bin6
303 prop_binSR_electron2018_bin3
304 prop_binCRTT_electron2017_bin9
305 prop_binCROS_muon2017_bin5
306 prop_binSR_muon2016M_bin3
307 prop_binCRTT_electron2016M_bin15
308 prop_binSR_electron2017_bin16
309 prop_binCROS_muon2017_bin3
310 prop_binCRTT_muon2016M_bin9
311 prop_binSR_electron2017_bin17_Triboson
312 prop_binCRTT_electron2018_bin16
313 prop_binCRTT_electron2016M_bin17
314 prop_binCRTT_electron2016M_bin13
315 prop_binCRTT_electron2016M_bin7
316 prop_binCRTT_electron2016M_bin9
317 QCDScale_Triboson
318 prop_binSR_muon2016M_bin16_TTTo2L2Nu
319 prop_binSR_muon2017_bin16_TVX
320 prop_binSR_electron2016M_bin4
321 prop_binSR_muon2018_bin9
322 prop_binCROS_electron2017_bin2
323 prop_binSR_muon2016M_bin16_ZZtoLep
324 prop_binCROS_electron2016M_bin14
325 prop_binSR_electron2016M_bin14
326 prop_binSR_muon2017_bin14_WpWpJJ_EWK
327 prop_binCRTT_muon2018_bin17
328 prop_binCRTT_electron2017_bin12
329 prop_binCROS_electron2016M_bin12
330 prop_binSR_electron2018_bin16_WpWpJJ_EWK
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



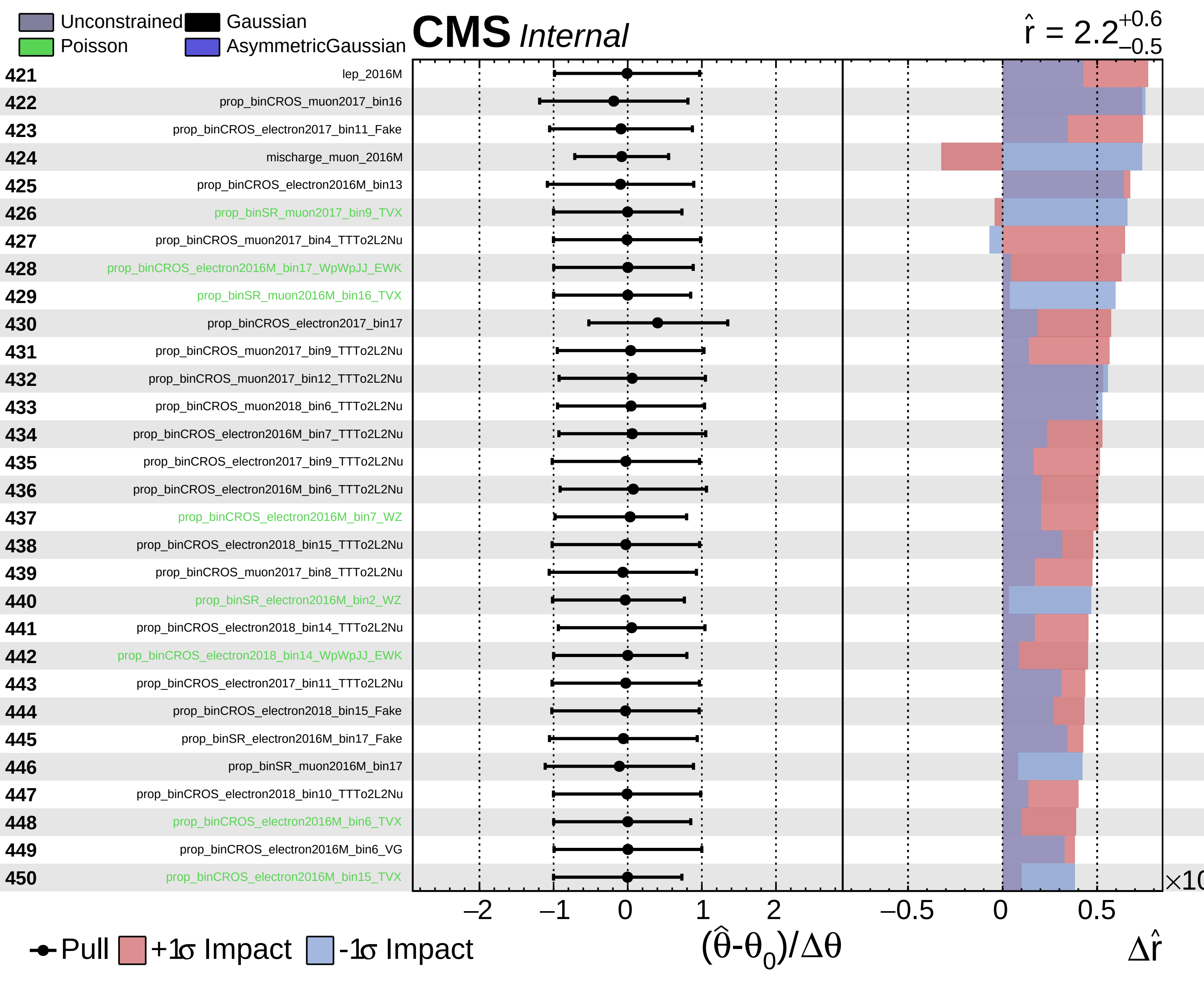
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.2^{+0.6}_{-0.5}$
331 prop_binCROS_electron2018_bin12
332 prop_binCROS_electron2018_bin4
333 prop_binCRTT_electron2017_bin13
334 prop_binCROS_muon2016M_bin6
335 prop_binCRTT_electron2017_bin14
336 prop_binSR_muon2017_bin10
337 prop_binSR_electron2016M_bin7
338 mischarge_electron_2018
339 prop_binCROS_electron2018_bin5
340 prop_binCROS_muon2017_bin12_Fake
341 prop_binSR_electron2018_bin13
342 prop_binCRTT_muon2018_bin14
343 prop_binCROS_muon2016M_bin7
344 jer_2016M
345 prop_binCRTT_electron2016M_bin14
346 prop_binCRTT_muon2016M_bin12
347 prop_binCROS_electron2017_bin14
348 prop_binCRTT_electron2017_bin15
349 prop_binSR_electron2017_bin15
350 prop_binSR_electron2016M_bin17_WpWpJJ_EWK
351 lep_2017
352 prop_binSR_muon2017_bin17
353 prop_binCROS_electron2017_bin10_TTTo2L2Nu
354 prop_binCRTT_electron2016M_bin16
355 prop_binCROS_electron2018_bin14_Fake
356 prop_binCROS_electron2018_bin16
357 QCDScale_TVX
358 prop_binSR_muon2017_bin16_WpWpJJ_EWK
359 prop_binCROS_electron2016M_bin16
360 prop_binCROS_muon2018_bin17
−2 −1 0 1 2
−0.002 0 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



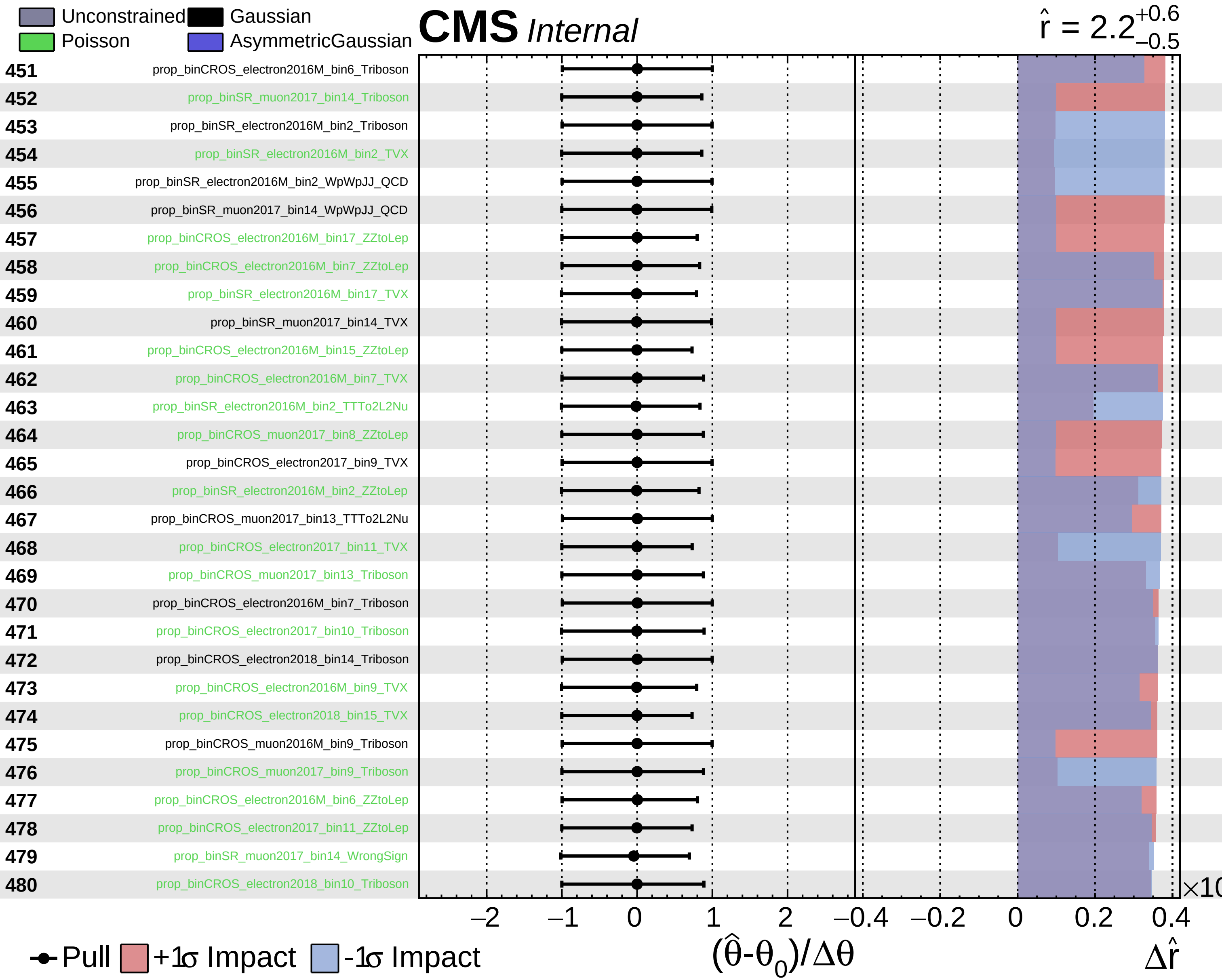
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.2^{+0.6}_{-0.5}$
361 prop_binSR_electron2018_bin16_TTTo2L2Nu
362 prop_binCROS_muon2016M_bin9_Fake
363 prop_binCROS_muon2017_bin7
364 prop_binSR_muon2017_bin16_WpWpJJ_QCD
365 prop_binSR_muon2016M_bin16_WpWpJJ_EWK
366 prop_binCROS_electron2016M_bin17_WrongSign
367 prop_binCROS_electron2016M_bin15_Fake
368 QCDScale_VG
369 prop_binSR_electron2017_bin17_TTTo2L2Nu
370 prop_binSR_electron2017_bin14
371 prop_binCROS_electron2016M_bin17_TTTo2L2Nu
372 prop_binSR_electron2016M_bin2_WrongSign
373 prop_binSR_muon2017_bin16_WrongSign
374 prop_binCROS_electron2018_bin13
375 prop_binSR_electron2017_bin17_WpWpJJ_EWK
376 prop_binCROS_muon2018_bin13
377 prop_binCROS_electron2016M_bin10
378 prop_binSR_muon2017_bin16_Triboson
379 prop_binCRTT_muon2016M_bin16
380 prop_binCRTT_electron2017_bin16
381 prop_binSR_electron2017_bin13_WpWpJJ_EWK
382 prop_binSR_muon2017_bin16_ZZtoLep
383 prop_binCROS_muon2017_bin9_Fake
384 mischarge_electron_2016M
385 prop_binCROS_electron2017_bin12
386 prop_binCRTT_muon2016M_bin15
387 prop_binCROS_muon2016M_bin14
388 QCDScale_ZZtoLep
389 prop_binSR_electron2018_bin16_TVX
390 prop_binCROS_electron2016M_bin9_Fake
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



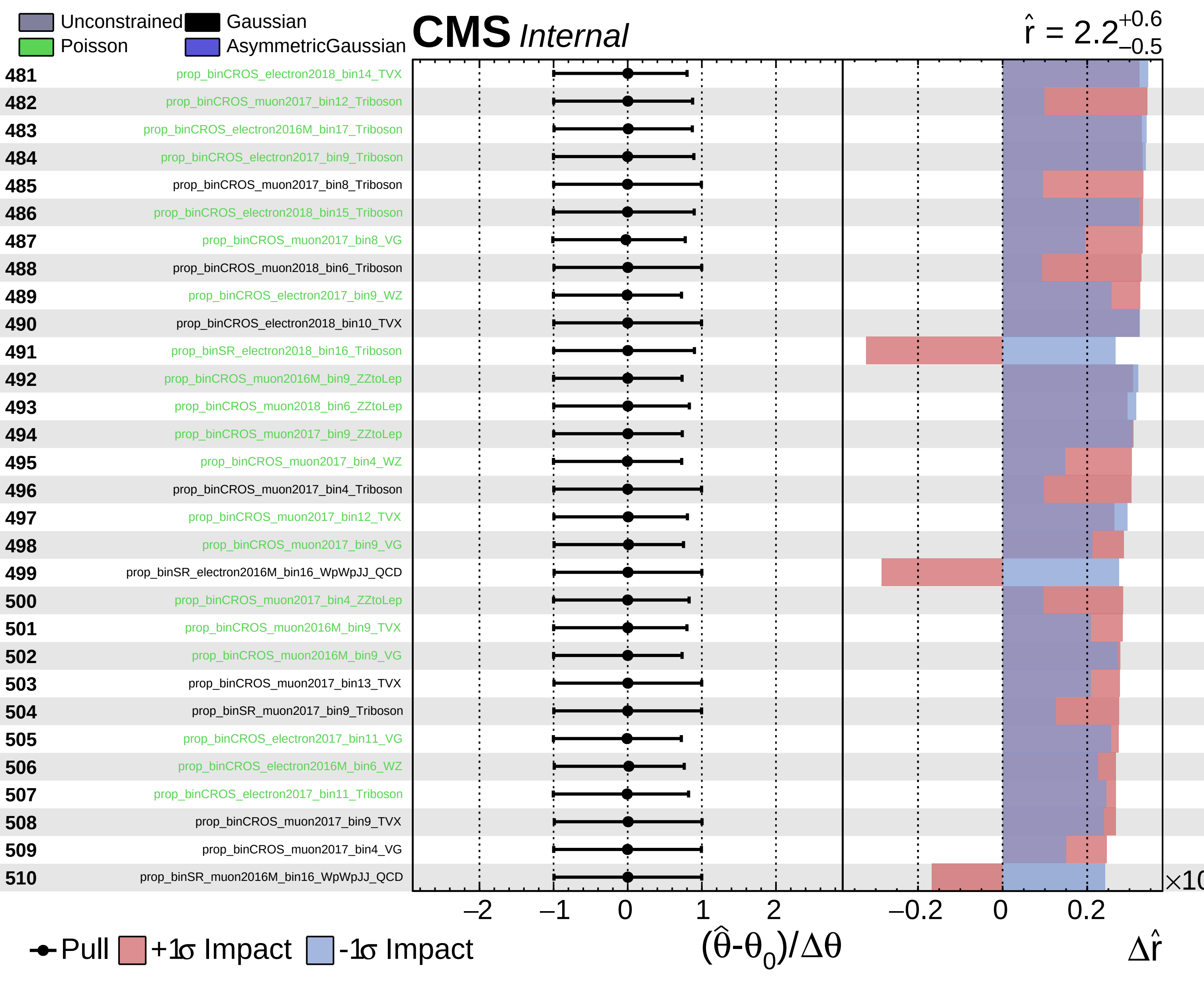
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.2^{+0.6}_{-0.5}$
391 prop_binSR_electron2016M_bin16_WpWpJJ_EWK
392 prop_binCROS_electron2016M_bin9_TTTo2L2Nu
393 prop_binCROS_muon2017_bin13_Fake
394 prop_binCROS_electron2017_bin15
395 lep_2018
396 ISR
397 prop_binCROS_muon2016M_bin15
398 prop_binCROS_muon2016M_bin13
399 prop_binCROS_muon2017_bin17
400 prop_binCROS_muon2017_bin15
401 prop_binCROS_electron2017_bin16
402 prop_binCRTT_muon2016M_bin17
403 prop_binCROS_muon2017_bin11
404 mischarge_electron_2017
405 prop_binSR_muon2017_bin9_WpWpJJ_EWK
406 prop_binSR_electron2017_bin13_TTTo2L2Nu
407 prop_binCROS_electron2018_bin15_VG
408 prop_binCRTT_electron2017_bin17
409 prop_binCROS_muon2016M_bin17
410 prop_binSR_muon2016M_bin15
411 prop_binCROS_muon2016M_bin16
412 prop_binCROS_muon2018_bin15
413 prop_binSR_muon2017_bin13
414 prop_binCROS_electron2016M_bin15_TTTo2L2Nu
415 prop_binSR_muon2016M_bin16_WrongSign
416 prop_binCROS_muon2016M_bin12
417 prop_binCROS_muon2018_bin16
418 prop_binSR_electron2016M_bin2_WpWpJJ_EWK
419 prop_binCROS_electron2018_bin17
420 prop_binCROS_muon2016M_bin9_TTTo2L2Nu
−2 −1 0 1 2
−0.001 0 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

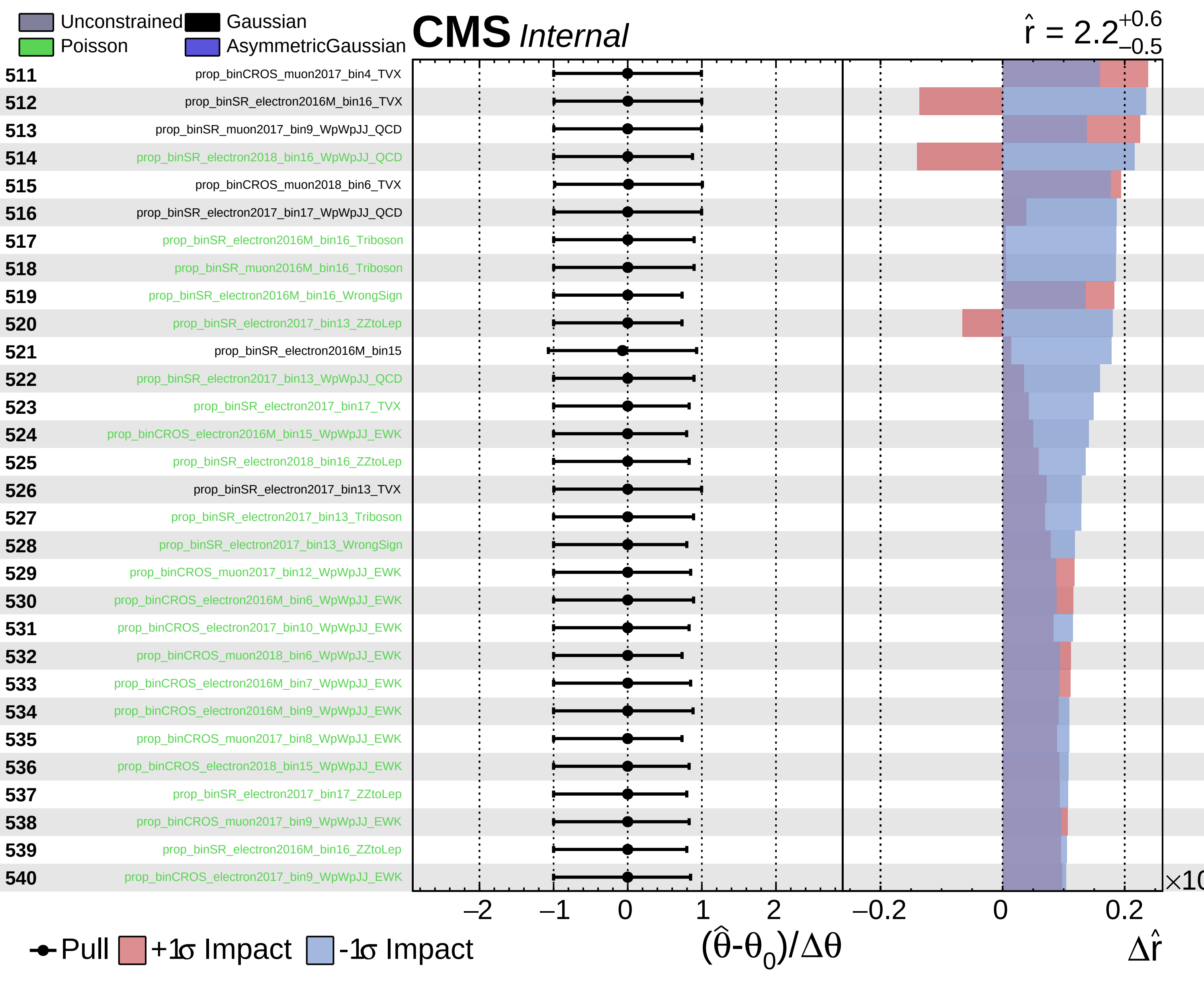


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.2^{+0.6}_{-0.5}$
421 lep_2016M
422 prop_binCROS_muon2017_bin16
423 prop_binCROS_electron2017_bin11_Fake
424 mischarge_muon_2016M
425 prop_binCROS_electron2016M_bin13
426 prop_binSR_muon2017_bin9_TVX
427 prop_binCROS_muon2017_bin4_TTTo2L2Nu
428 prop_binCROS_electron2016M_bin17_WpWpJJ_EWK
429 prop_binSR_muon2016M_bin16_TVX
430 prop_binCROS_electron2017_bin17
431 prop_binCROS_muon2017_bin9_TTTo2L2Nu
432 prop_binCROS_muon2017_bin12_TTTo2L2Nu
433 prop_binCROS_muon2018_bin6_TTTo2L2Nu
434 prop_binCROS_electron2016M_bin7_TTTo2L2Nu
435 prop_binCROS_electron2017_bin9_TTTo2L2Nu
436 prop_binCROS_electron2016M_bin6_TTTo2L2Nu
437 prop_binCROS_electron2016M_bin7_WZ
438 prop_binCROS_electron2018_bin15_TTTo2L2Nu
439 prop_binCROS_muon2017_bin8_TTTo2L2Nu
440 prop_binSR_electron2016M_bin2_WZ
441 prop_binCROS_electron2018_bin14_TTTo2L2Nu
442 prop_binCROS_electron2018_bin14_WpWpJJ_EWK
443 prop_binCROS_electron2017_bin11_TTTo2L2Nu
444 prop_binCROS_electron2018_bin15_Fake
445 prop_binSR_electron2016M_bin17_Fake
446 prop_binSR_muon2016M_bin17
447 prop_binCROS_electron2018_bin10_TTTo2L2Nu
448 prop_binCROS_electron2016M_bin6_TVX
449 prop_binCROS_electron2016M_bin6_VG
450 prop_binCROS_electron2016M_bin15_TVX
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

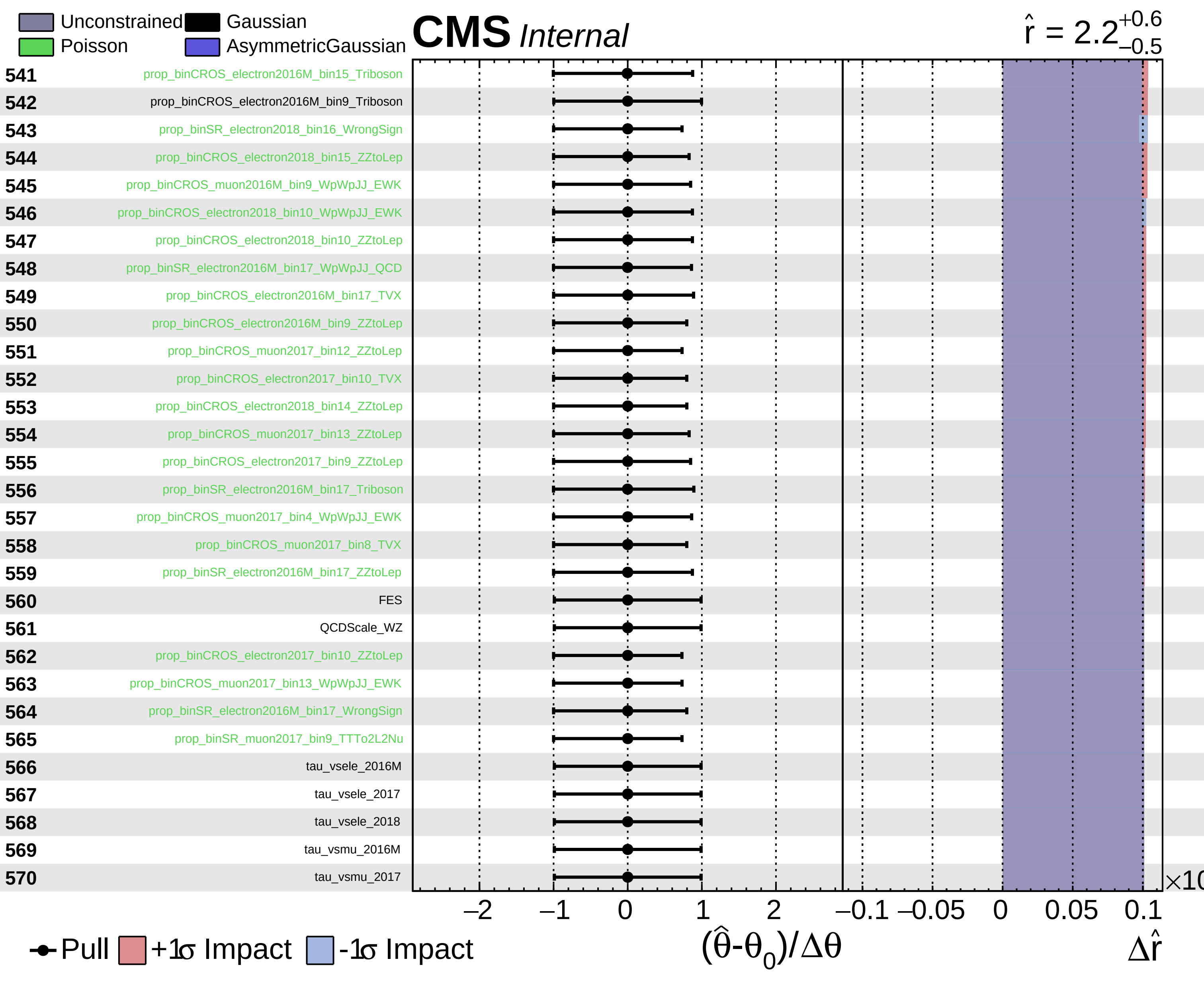


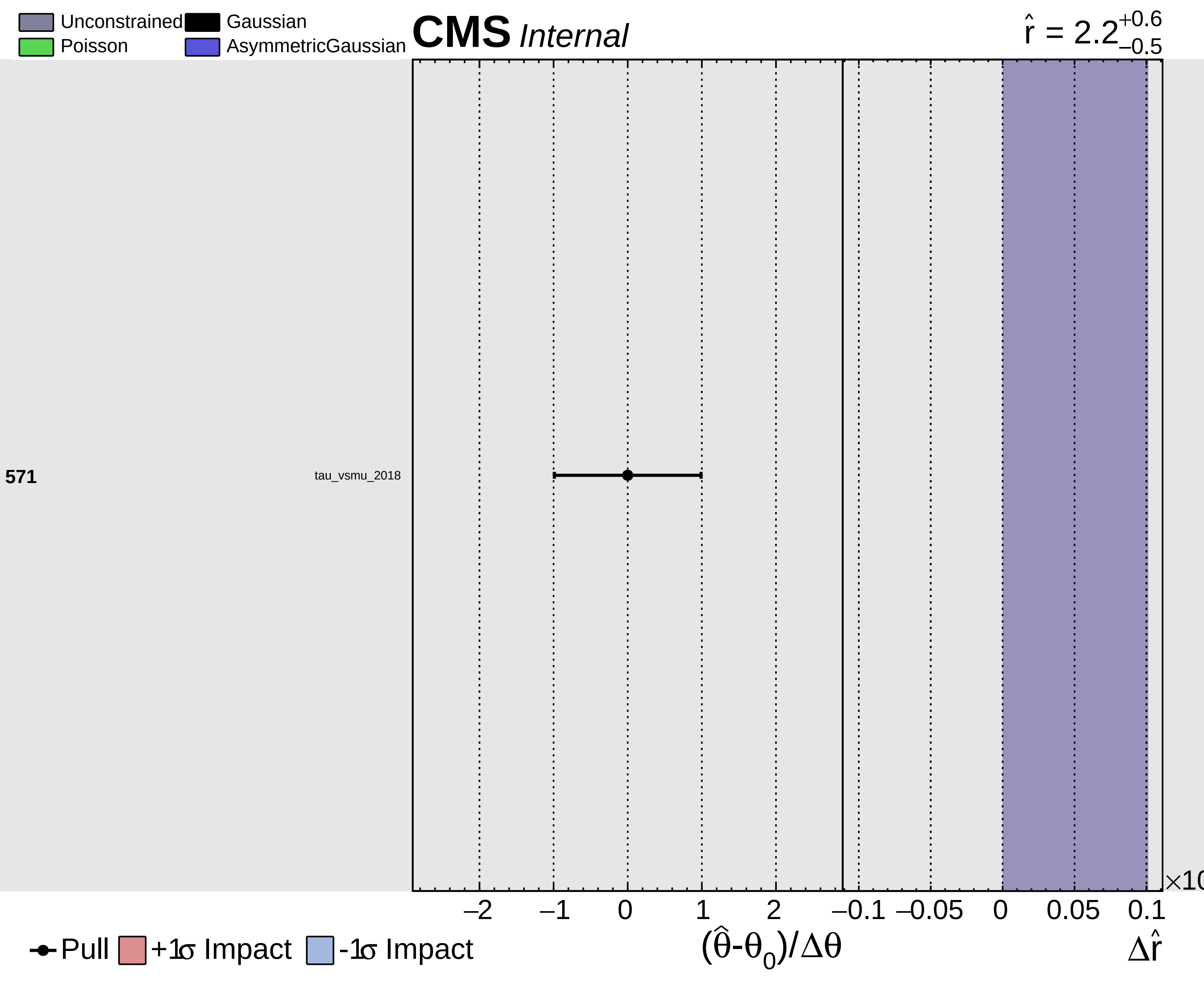
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.2^{+0.6}_{-0.5}$
451 prop_binCROS_electron2016M_bin6_Triboson
452 prop_binSR_muon2017_bin14_Triboson
453 prop_binSR_electron2016M_bin2_Triboson
454 prop_binSR_electron2016M_bin2_TVX
455 prop_binSR_electron2016M_bin2_WpWpJJ_QCD
456 prop_binSR_muon2017_bin14_WpWpJJ_QCD
457 prop_binCROS_electron2016M_bin17_ZZtoLep
458 prop_binCROS_electron2016M_bin7_ZZtoLep
459 prop_binSR_electron2016M_bin17_TVX
460 prop_binSR_muon2017_bin14_TVX
461 prop_binCROS_electron2016M_bin15_ZZtoLep
462 prop_binCROS_electron2016M_bin7_TVX
463 prop_binSR_electron2016M_bin2_TTTo2L2Nu
464 prop_binCROS_muon2017_bin8_ZZtoLep
465 prop_binCROS_electron2017_bin9_TVX
466 prop_binSR_electron2016M_bin2_ZZtoLep
467 prop_binCROS_muon2017_bin13_TTTo2L2Nu
468 prop_binCROS_electron2017_bin11_TVX
469 prop_binCROS_muon2017_bin13_Triboson
470 prop_binCROS_electron2016M_bin7_Triboson
471 prop_binCROS_electron2017_bin10_Triboson
472 prop_binCROS_electron2018_bin14_Triboson
473 prop_binCROS_electron2016M_bin9_TVX
474 prop_binCROS_electron2018_bin15_TVX
475 prop_binCROS_muon2016M_bin9_Triboson
476 prop_binCROS_muon2017_bin9_Triboson
477 prop_binCROS_electron2016M_bin6_ZZtoLep
478 prop_binCROS_electron2017_bin11_ZZtoLep
479 prop_binSR_muon2017_bin14_WrongSign
480 prop_binCROS_electron2018_bin10_Triboson
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 2.2 +0.6 −0.5
481 prop_binCROS_electron2018_bin14_TVX
482 prop_binCROS_muon2017_bin12_Triboson
483 prop_binCROS_electron2016M_bin17_Triboson
484 prop_binCROS_electron2017_bin9_Triboson
485 prop_binCROS_muon2017_bin8_Triboson
486 prop_binCROS_electron2018_bin15_Triboson
487 prop_binCROS_muon2017_bin8_VG
488 prop_binCROS_muon2018_bin6_Triboson
489 prop_binCROS_electron2017_bin9_WZ
490 prop_binCROS_electron2018_bin10_TVX
491 prop_binSR_electron2018_bin16_Triboson
492 prop_binCROS_muon2016M_bin9_ZZtoLep
493 prop_binCROS_muon2018_bin6_ZZtoLep
494 prop_binCROS_muon2017_bin9_ZZtoLep
495 prop_binCROS_muon2017_bin4_WZ
496 prop_binCROS_muon2017_bin4_Triboson
497 prop_binCROS_muon2017_bin12_TVX
498 prop_binCROS_muon2017_bin9_VG
499 prop_binSR_electron2016M_bin16_WpWpJJ_QCD
500 prop_binCROS_muon2017_bin4_ZZtoLep
501 prop_binCROS_muon2016M_bin9_TVX
502 prop_binCROS_muon2016M_bin9_VG
503 prop_binCROS_muon2017_bin13_TVX
504 prop_binSR_muon2017_bin9_Triboson
505 prop_binCROS_electron2017_bin11_VG
506 prop_binCROS_electron2016M_bin6_WZ
507 prop_binCROS_electron2017_bin11_Triboson
508 prop_binCROS_muon2017_bin9_TVX
509 prop_binCROS_muon2017_bin4_VG
510 prop_binSR_muon2016M_bin16_WpWpJJ_QCD
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.2^{+0.6}_{-0.5}$
511 prop_binCROS_muon2017_bin4_TVX
512 prop_binSR_electron2016M_bin16_TVX
513 prop_binSR_muon2017_bin9_WpWpJJ_QCD
514 prop_binSR_electron2018_bin16_WpWpJJ_QCD
515 prop_binCROS_muon2018_bin6_TVX
516 prop_binSR_electron2017_bin17_WpWpJJ_QCD
517 prop_binSR_electron2016M_bin16_Triboson
518 prop_binSR_muon2016M_bin16_Triboson
519 prop_binSR_electron2016M_bin16_WrongSign
520 prop_binSR_electron2017_bin13_ZZtoLep
521 prop_binSR_electron2016M_bin15
522 prop_binSR_electron2017_bin13_WpWpJJ_QCD
523 prop_binSR_electron2017_bin17_TVX
524 prop_binCROS_electron2016M_bin15_WpWpJJ_EWK
525 prop_binSR_electron2018_bin16_ZZtoLep
526 prop_binSR_electron2017_bin13_TVX
527 prop_binSR_electron2017_bin13_Triboson
528 prop_binSR_electron2017_bin13_WrongSign
529 prop_binCROS_muon2017_bin12_WpWpJJ_EWK
530 prop_binCROS_electron2016M_bin6_WpWpJJ_EWK
531 prop_binCROS_electron2017_bin10_WpWpJJ_EWK
532 prop_binCROS_muon2018_bin6_WpWpJJ_EWK
533 prop_binCROS_electron2016M_bin7_WpWpJJ_EWK
534 prop_binCROS_electron2016M_bin9_WpWpJJ_EWK
535 prop_binCROS_muon2017_bin8_WpWpJJ_EWK
536 prop_binCROS_electron2018_bin15_WpWpJJ_EWK
537 prop_binSR_electron2017_bin17_ZZtoLep
538 prop_binCROS_muon2017_bin9_WpWpJJ_EWK
539 prop_binSR_electron2016M_bin16_ZZtoLep
540 prop_binCROS_electron2017_bin9_WpWpJJ_EWK
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.2^{+0.6}_{-0.5}$
541 prop_binCROS_electron2016M_bin15_Triboson
542 prop_binCROS_electron2016M_bin9_Triboson
543 prop_binSR_electron2018_bin16_WrongSign
544 prop_binCROS_electron2018_bin15_ZZtoLep
545 prop_binCROS_muon2016M_bin9_WpWpJJ_EWK
546 prop_binCROS_electron2018_bin10_WpWpJJ_EWK
547 prop_binCROS_electron2018_bin10_ZZtoLep
548 prop_binSR_electron2016M_bin17_WpWpJJ_QCD
549 prop_binCROS_electron2016M_bin17_TVX
550 prop_binCROS_electron2016M_bin9_ZZtoLep
551 prop_binCROS_muon2017_bin12_ZZtoLep
552 prop_binCROS_electron2017_bin10_TVX
553 prop_binCROS_electron2018_bin14_ZZtoLep
554 prop_binCROS_muon2017_bin13_ZZtoLep
555 prop_binCROS_electron2017_bin9_ZZtoLep
556 prop_binSR_electron2016M_bin17_Triboson
557 prop_binCROS_muon2017_bin4_WpWpJJ_EWK
558 prop_binCROS_muon2017_bin8_TVX
559 prop_binSR_electron2016M_bin17_ZZtoLep
560 FES
561 QCDScale_WZ
562 prop_binCROS_electron2017_bin10_ZZtoLep
563 prop_binCROS_muon2017_bin13_WpWpJJ_EWK
564 prop_binSR_electron2016M_bin17_WrongSign
565 prop_binSR_muon2017_bin9_TTTo2L2Nu
566 tau_vsele_2016M
567 tau_vsele_2017
568 tau_vsele_2018
569 tau_vsmu_2016M
570 tau_vsmu_2017
−2 −1 0 1 2
−0.1 −0.05 0 0.05 0.1
×10
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$
Pull
+1σ Impact
-1σ Impact



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.2^{+0.6}_{-0.5}$
571
tau_vsmu_2018
−2 −1 0 1 2
−0.1 −0.05 0 0.05 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$